

Xiaopeng Zhang

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EDUCATION BACKGROUND

Rutgers University, New Brunswick <i>Ph.D in Computer Science, advised by Prof. Jingjin Yu</i>	New Brunswick, US <i>August 2026 - 2031</i>
Toyota Technological Institute at Chicago <i>Visiting Student, advised by Prof. Matthew R. Walter</i>	Chicago, US <i>March 2025 - April 2026</i>
Data Science Institute (DSI), University of Chicago <i>M.S. in Data Science GPA: 3.85/4.00</i>	Chicago, US <i>Sept 2024 - June 2026</i>
Institute for Advanced Study (IAS), Shenzhen University <i>B.S. in Mathematics and Applied Mathematics, Graduate with Honor</i> Cumulative GPA: 4.21/4.5 (Weighted Average Score: 91.8) Ranking: Top 1 (1/17 in mathematics, 1/57 in IAS)	Shenzhen, China <i>Sept 2020 - June 2024</i>

AWARDS & ACHIEVEMENTS

• National Scholarship (TOP 0.2% nationwide), Chinese Ministry of Education	2022-2023
• Outstanding Graduate with Honor (top 1%)	2024
• National First Prize in the 14th National Mathematics Competition (36th nationwide)	2023
• National Second Prize , China Undergraduate Mathematical Contest in Modeling	2023
• First Prize in Guangdong Region , 13&14th National Mathematics Competition	2021, 2022
• Outstanding Innovative Talent (First Prize)	2020-2023
• Star of Learning (First Prize) - Top 1 in IAS	2021, 2022, 2023
• Discipline Scholarship (Top 0.5% in National College Entrance Examination, 642/750)	2020

RESEARCH EXPERIENCE

Robot Intelligence through Perception Laboratory (RIPL), TTIC	Advisor: Matthew R. Walter
1. StackGen3D: Stable 3D Block Stack Generation from Multiview Silhouettes	Mar 2025 - April 2026
• Accepted to the IROS 2026 Workshop on Creativity in Assembly (co-first author)	
• Curated stable training data with a physics-informed GNN and MuJoCo verification	
• Generated block poses from silhouettes and a block inventory via conditional Flow Matching	
• Designed simulator-free AABB-based inference , raising stability from 45.3% to 72.4%	
2. Assist When You Need	November 2025 - Present
• Implemented Flow Matching for preserving user safe commands deemed by the model in shared autonomy rather than adding residual correction, respecting user intent even if suboptimal	

TEACHING EXPERIENCE

Teaching Assistant, Advanced Calculus , IAS, SZU	Sept 2022 - Jan 2023
Teaching Assistant, Advanced Algebra , IAS, SZU	Sept 2023 - Jan 2024

STANDARDIZED TESTS

- **GRE Mathematics Subject Test:** 900 (93%)

SKILLS

- **Programming:** Python, MATLAB, R
- **Coding Frameworks:** PyTorch, TensorFlow, ROS
- **Tools/Platform:** Git, Docker, Google Cloud Platform, Huggingface
- **Research Areas:** Robotics, Diffusion Models, Generative Models
- **Languages:** Chinese (Native), English (Fluent)